UNIVERSITY OF LONDON

CM3060

BSc EXAMINATION

COMPUTER SCIENCE

Natural Language Processing

Release date: Tuesday 21 March 2023 at 12:00 midday Greenwich Mean Time

Submission date: Wednesday 22 March 2023 by 12:00 midday Greenwich Mean Time

Time allowed: 24 hours to submit

INSTRUCTIONS TO CANDIDATES:

Section A of this assessment paper consists of a set of **TEN** Multiple Choice Questions (MCQs) which you will take separately from this paper. You should attempt to answer **ALL** the questions in Section A. The maximum mark for Section A is **40**.

Section A will be completed online on the VLE. You may choose to access the MCQs at any time following the release of the paper, but once you have accessed the MCQs you must submit your answers before the deadline or within **4 hours** of starting whichever occurs first.

Section B of this assessment paper is an online assessment to be completed within the same 24-hour window as Section A. We anticipate that approximately **1 hour** is sufficient for you to answer Section B. Candidates must answer **TWO** out of the THREE questions in Section B. The maximum mark for Section B is **60**.

Calculators are not permitted in this examination. Credit will only be given if all workings are shown.

You should complete **Section B** of this paper and submit your answers as **one document**, if possible, in Microsoft Word or a PDF to the appropriate area on the VLE. Each file uploaded must be accompanied by a coversheet containing your **candidate number**. In addition, your answers must have your candidate number written clearly at the top of the page before you upload your work. Do not write your name anywhere in your answers.

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SECTION A

Candidates should answer the **TEN** Multiple Choice Questions (MCQs) quiz, **Question 1** in Section A on the VLE.

SECTION B

Candidates should answer any **TWO** questions from Section B.

Question 2

(a) What are sequence-to-sequence models in NLP? Explain briefly how they work. [4]

(b) How might the use of an attention mechanism improve the above process? [4]

(c) In a 100-million-word corpus we find the following counts of unigrams and bigrams:

my	876,543
thoughts	87,654
carpets	21
my thoughts	378
my carpets	0

Estimate the probabilities P(thoughts) and P(thoughts | my) using Maximum Likelihood estimation (without smoothing). Express your answer as a fraction. [5]

(d) Estimate the bigram probability P(carpets | my) using Maximum Likelihood Estimation and Laplace smoothing. Assume that the vocabulary consists of 1,234,567 unique words. Express your answer as a fraction. [5]

(e) Estimate the bigram probability P(carpets | my) using Maximum Likelihood Estimation and add-k smoothing with k = 0.2. Assume that the vocabulary consists of 1,234,567 unique words. Express your answer as a fraction. [5]

(f) Given the following training data, calculate the likelihood probabilities for each word given each POS tag.

- plane/NN flying/VBG is/VBZ stressful/JJ
- flying/VBG planes/NNS are/VBP dangerous/JJ
- I/PRP saw/VBD Lauren/NNP flying/VBG planes/NNS
- She/PRP planes/NNS doors/NNS

[7]

Question 3

(a) Describe the process of sentence tokenization. What challenges might you encounter with a sentence like “Prof. Russell-Rose uses TF.IDF when parsing data from Which? reports (but only occasionally :)”

[7]

(b) Define word tokenization. What challenges might you encounter in tokenizing the following text? To what extent are these challenges unique to English? [8]

```
Mr. Martin O’Neill was the Republic of Ireland’s manager for the
period 2013-2018. They qualified for UEFA Euro 2016 after
defeating Bosnia and Herzegovina 3–1 in the play-offs. Overall:
it didn’t work out...
```

(c) Identify any named entities and their types in the example given. What are the main challenges in doing so?

[4]

(d) Define the bag-of-words model and briefly explain the processing steps you might take to represent the following sentences using such a model [5]:

```
Lauren likes to watch football, especially Liverpool.
Matthew likes football too.
```

(e) Suppose that you are building a model to identify topics in a corpus of 10 million movie reviews. What techniques could you use to reduce the dimensionality of the data? How might you use TF.IDF to represent the documents? [6]

Question 4

(a) A sentiment analyzer that predicts if a product review is positive, negative or neutral produces the following confusion matrix:

		True values		
		Positive	Neutral	Negative
Predicted values	Positive	20	3	2
	Neutral	4	18	2
	Negative	3	1	7

Calculate the following:

(i) Precision and recall for each class [6]

(ii) Micro average precision and macro average precision. Why do they differ? [4]

(iii) F measure for each class [6]

(b) You are designing a text classifier to identify fraudulent insurance claims and your initial prototype returns an accuracy of 97%. Why might this figure be misleading? Suggest two strategies for dealing with this situation [5]

(c) What techniques might you use to deal with typographic errors in search queries? Suggest one method based on distributional techniques and another based on a more rule-based or algorithmic approach. [4]

(d) Describe or draw the correct parse tree for the most natural interpretation of the following English sentence: ‘She took the lesson to heart’.[5]

END OF PAPER